Quantification: 548_hyp_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	24
interior lacuna count	18
frame-edge lacuna count	6

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2579.22	1362.5	2390.43	px^2
major_axis_length_px	lacuna shape	79.9015	66.1671	39.0496	px
minor_axis_length_px	lacuna shape	39.6826	29.6555	20.6373	px
aspect_ratio	lacuna shape	2.152	2.0364	0.6587	unitless
eccentricity	lacuna shape	0.8481	0.8711	0.0963	unitless
solidity	lacuna shape	0.802	0.8096	0.0866	unitless
roots_count	headline	8.7778	7.5	5.0823	count
ring_length_r30_px	headline	277.944	263	132.551	px
ring_length_r60_px	reported	511.611	492.5	254.055	px
owned_length_px	ownership-dependent	689.782	457.253	579.334	px
edge_count	ownership-dependent	37.1111	22.5	29.5234	count
mean_edge_length_px	ownership-dependent	18.1345	17.2541	2.8094	px
perimeter_px	normalised	276.922	229.693	173.965	px
ring_area_r30_px2	normalised	8628.72	7185	3524.72	px^2
ring_area_r60_px2	normalised	17290.1	15932.5	7468.53	px^2
in_frame_fraction_r30	normalised	0.9872	1	0.0372	unitless
in_frame_fraction_r60	normalised	0.9584	1	0.0769	unitless
ring_density_r30	normalised	0.0320118	0.0315463	0.00695829	px^-1
ring_density_r60	normalised	0.0292802	0.0283762	0.00542875	px^-1
roots_per_100px_perimeter	normalised	3.5428	3.2008	1.5556	per 100 px
ring_attached_length_r30_px	network v2	244.5	244	124.603	px
ring_attached_length_r60_px	network v2	371.389	317	206.609	px
ring_length_w_r30_px	network v2	315.24	302.472	150.847	px
ring_length_w_r60_px	network v2	583.256	563.644	293.817	px
sholl_crossings_r10	network v2	6.6667	6.5	3.1993	count
sholl_crossings_r20	network v2	7.1111	7	3.496	count
sholl_crossings_r30	network v2	6.3889	6	3.4324	count
loop_count	loops, report only	1.7778	1	2.7128	count
loop_edge_share	loops, report only	0.1042	0.0882	0.1169	share
ring_width_mean_r30_px	width	6.907	6.9147	1.2302	px

Per field
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measure	value	units
analysed_area_px2	991908	px^2
total_skeleton_length_px	25169	px
canalicular_length_density_per_px	0.0253743	px^-1
junction_count	518	count
junction_density_per_px2	0.00052223	px^-2
lacuna_count	24	count

lacunae_per_px2	2.42e-05	px^-2
skeleton_component_count	430	count
mean_component_length_px	58.5326	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0288695	px^-1
field_density_without_flagged_per_px	0.0240473	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	541	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0253743	px^-1
field_length_density_with_unattached_per_px	0.0259197	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	45	count
loop_edge_share	0.11	share
edge_count	1455	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4382	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21655	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1331	68.1738	43.3369	1.5731	0.772	0.5571	0.8707	22.9699	562.642	True
2	983	77.6846	19.2927	4.0266	0.9687	0.7097	-0.7276	26.3245	837.443	True
3	1345	60.6773	30.0142	2.0216	0.8691	0.8739	0.6556	34.9123	469.557	False
4	5304	115.818	61.0253	1.8979	0.8499	0.8497	-0.5279	76.0664	899.181	False
5	1100	56.8071	27.1347	2.0935	0.8785	0.8035	0.1617	61.0436	700.865	False
6	573	40.0083	19.0937	2.0954	0.8788	0.8788	-1.5045	203.53	964.757	False
7	589	35.3245	25.9374	1.3619	0.6789	0.8158	0.6135	217.115	155.742	False
8	1020	65.1958	29.2967	2.2254	0.8933	0.5543	0.7245	258.288	921.093	False
9	508	48.1299	13.5874	3.5423	0.9593	0.9153	0.8262	258.841	719	False
10	2999	76.2507	53.7089	1.4197	0.7098	0.8844	-0.1093	288.357	25.6549	True
11	8181	176.425	74.8525	2.357	0.9055	0.7394	-0.9469	333.681	384.601	False
12	772	43.3165	23.8026	1.8198	0.8355	0.8863	1.0039	324.626	799.803	False
13	843	67.1384	19.7863	3.3932	0.9556	0.7938	-0.8751	303.813	882.996	False
14	5780	125.902	64.4212	1.9544	0.8592	0.8211	0.4178	372.596	722.104	False
15	680	44.3797	22.5335	1.9695	0.8615	0.8543	-0.0423	380.915	8.0779	True
16	3628	78.9766	63.2544	1.2486	0.5988	0.8761	0.6151	443.631	464.428	False
17	561	40.1918	19.6867	2.0416	0.8718	0.8738	0.4324	421.501	794.918	False
18	1526	56.8255	39.8642	1.4255	0.7126	0.7592	0.9788	480.14	672.478	False
19	2634	99.9739	44.742	2.2345	0.8943	0.697	1.1806	489.293	203.02	False
20	5796	118.655	71.2141	1.6662	0.7999	0.7445	0.2563	543.606	765.066	False
21	2266	83.3012	36.8142	2.2627	0.897	0.8389	0.7457	720.831	996.501	True
22	1380	84.7713	25.484	3.3264	0.9537	0.7908	-1.481	787.319	890.853	False

23	1983	87.7473	31.9283	2.7483	0.9315	0.8015	-1.3962	877.209	988.193	True
24	4886	124.09	61.0897	2.0313	0.8704	0.7632	0.9159	965.955	722.24	False

Network per lacuna

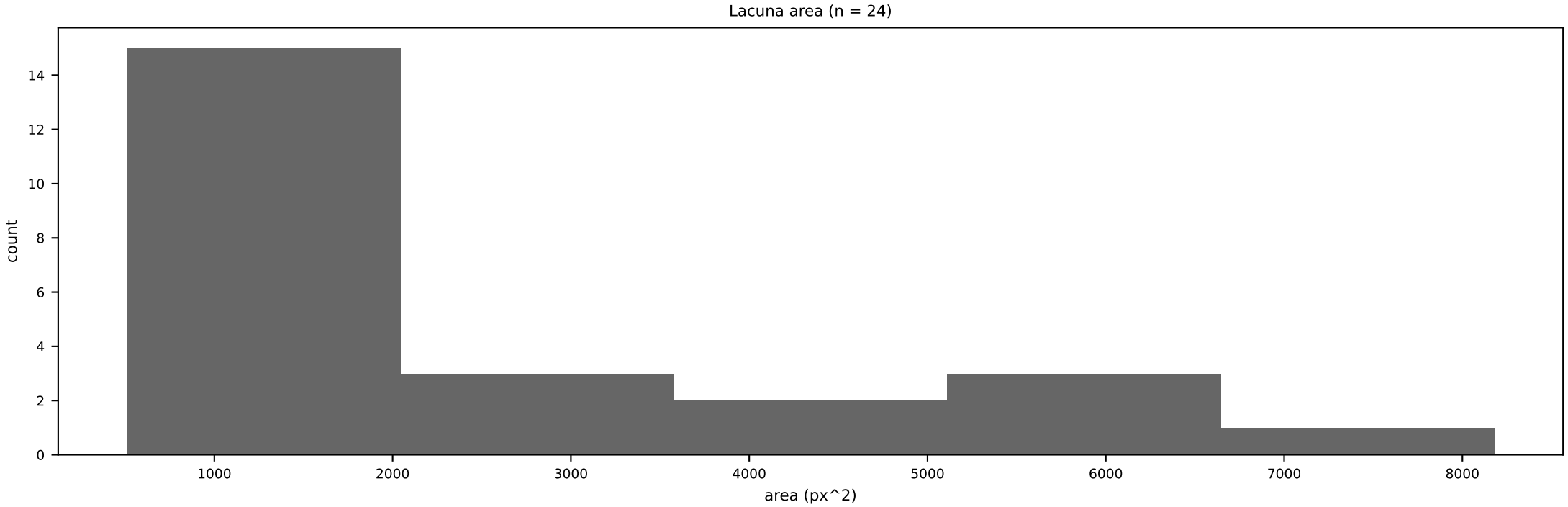
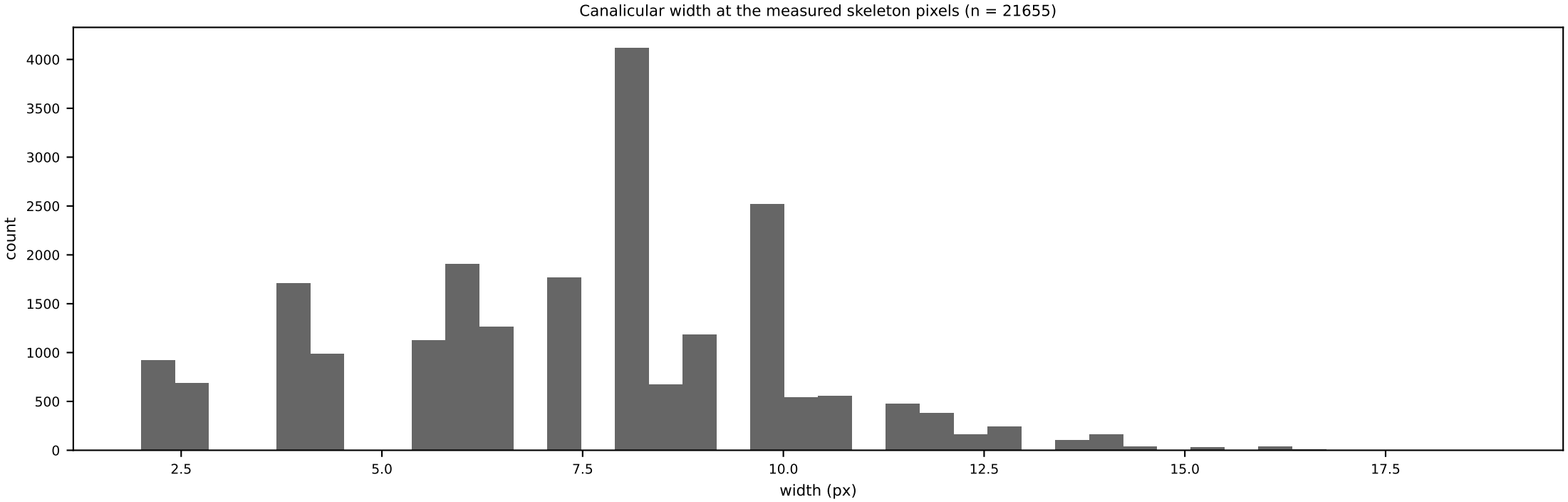
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	203	364	593.186	19	31.2203	10.1383
2	6	201	376	486.475	34	14.3081	6.7471
3	4	237	451	414.576	17	24.3868	6.6843
4	9	356	526	499.931	23	21.7361	8.2994
5	9	316	541	559.588	34	16.4585	6.9559
6	6	133	232	167.711	10	16.7711	6.3009
7	7	226	491	359.907	18	19.9948	6.4606
8	12	266	442	317.907	19	16.7319	7.2483
9	7	168	328	286.865	18	15.9369	6.2019
10	5	207	361	944.021	53	17.8117	7.5914
11	24	658	1200	1942.8	103	18.8621	6.7801
12	3	171	190	193.066	13	14.8512	6.9034
13	7	151	354	319.563	19	16.8191	6.4736
14	7	213	364	323.522	19	17.0275	8.0629
15	2	60	135	24.7279	1	24.7279	7.4395
16	12	355	734	638.316	40	15.9579	6.7865
17	2	89	177	1387.51	62	22.3792	2.8258
18	13	284	494	656.085	34	19.2966	7.4281
19	11	423	910	640.328	34	18.8332	7.5163
20	13	395	673	1899.03	96	19.7816	8.5257
21	5	262	440	165.225	9	18.3584	7.2195
22	4	260	521	1520.82	87	17.4807	7.947
23	7	252	529	571.872	26	21.9951	8.3853
24	8	302	581	288.551	22	13.116	6.926

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	299.12	6629	11618	0.7226	0.6609	0.030623	0.0313307	2.0059
2	191.238	5199	8502	0.7613	0.6818	0.0386613	0.0442249	3.1375
3	185.238	6495	12887	0.8654	0.7462	0.0364896	0.0349965	2.1594
4	355.505	11178	20871	0.996	0.8914	0.0318483	0.0252024	2.5316
5	180.894	7279	16151	1	0.8909	0.0434126	0.0334964	4.9753
6	108.811	5247	10269	1	0.9129	0.0253478	0.0225923	5.5141
7	130.368	5996	17021	1	1	0.0376918	0.0288467	5.3694
8	247.806	7003	13467	1	1	0.0379837	0.032821	4.8425
9	111.539	5551	12005	1	1	0.0302648	0.027322	6.2758
10	300.635	6306	12787	0.711	0.6516	0.0328259	0.0282318	1.6631
11	732.927	17463	35109	1	1	0.0376797	0.0341793	3.2745
12	129.095	5473	7747	1	1	0.0312443	0.0245256	2.3239
13	153.238	5367	9216	1	1	0.0281349	0.0384115	4.5681
14	413.245	10985	15714	1	1	0.0193901	0.0231641	1.6939
15	129.154	3191	7077	0.5819	0.5549	0.0188029	0.0190759	1.5485
16	300.433	10348	22315	1	1	0.0343062	0.0328927	3.9942
17	106.569	4649	8339	1	1	0.0191439	0.0212256	1.8767
18	246.25	7091	13595	1	1	0.0400508	0.0363369	5.2792
19	351.747	10779	26637	1	1	0.039243	0.034163	3.1272
20	533.985	13725	25712	1	1	0.0287796	0.0261746	2.4345
21	256.108	6631	13116	0.7153	0.654	0.0395114	0.0335468	1.9523
22	213.137	8610	18670	1	1	0.0301974	0.0279057	1.8767
23	256.551	6858	14534	0.748	0.6875	0.0367454	0.0363974	2.7285

24 483.813 12078 25496 0.909 0.8104 0.0250041 0.0227879 1.6535

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	170	268	230.894	406.149	8	7	6
2	181	282	227.338	432.262	6	3	5
3	232	269	269.622	504.09	6	5	4
4	301	454	400.363	589.73	8	13	9
5	273	427	357.492	606.931	7	9	7
6	121	164	146.225	256.894	2	3	3
7	178	325	253.309	556.073	5	5	8
8	250	309	301.936	505.588	7	7	5
9	148	220	187.368	367.907	3	6	4
10	148	221	241.522	422.103	6	6	3
11	632	1022	753.097	1397.51	16	15	13
12	150	151	198.894	219.037	6	5	2
13	108	270	178.581	421.789	3	2	3
14	183	250	236.581	402.806	10	7	3
15	50	78	63.2132	145.74	1	1	1
16	280	433	392.321	819.34	7	9	11
17	89	151	103.083	206.823	4	2	2
18	284	494	326.25	571.215	5	6	5
19	302	482	474.718	1040.8	9	9	11
20	382	577	455.132	767.068	9	11	7
21	140	232	290.137	492.647	5	5	10
22	249	424	303.007	608.127	6	7	10
23	218	418	286.693	610.571	6	6	9
24	239	263	336.35	656.884	7	7	8

548_hyp_HP_z20c1-2: distributions, pre-validation, px